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Structural fuses configured to yield in tension and compression and structures including the same

Abstract

Embodiments are directed to structural fuses, structures including the same, and methods of using and forming the same. In an embodiment, a structural fuse is disclosed. The structural fuse includes a first attachment region defining one or more first bolt holes, a second attachment region defining one or more second bolt holes, and a yielding region between the first attachment region and the second attachment region. The yielding region defines one or more slots exhibiting an elongated shape.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
263426	12/1881	O'Brien	52/651.1	B60R 13/0206
1727293	12/1928	Larkin	403/173	E04B 1/2403
1744600	12/1929	Wilcox	403/230	E04B 1/2403
2106084	12/1937	Coddington	52/655.1	E04C 3/08
2114901	12/1937	Henderson	52/837	E04B 1/944
2114902	12/1937	Henderson	52/838	E04B 1/2403
3106270	12/1962	Lichty	52/301	E04B 1/2403
3716957	12/1972	Bernardi	52/837	E04B 1/5812
4014089	12/1976	Sato	228/140	E04B 1/2403
5148642	12/1991	Plumier	52/638	E04H 9/0237
5595040	12/1996	Chen	52/837	E04H 9/02
5660017	12/1996	Houghton	52/283	E04H 9/024
6012256	12/1999	Aschheim	52/573.1	E04C 3/086
6059482	12/1999	Beauvoir	403/187	F16B 9/058
6138427	12/1999	Houghton	52/655.1	E04B 1/2403
6516583	12/2002	Houghton	52/638	E04B 1/24
7497054	12/2008	Takeuchi	403/403	E04B 1/2403
7712266	12/2009	Sarkisian	52/573.1	E04H 9/0237
8615948	12/2012	Underkofler	52/506.07	E04B 9/067
8800239	12/2013	Yang	403/403	E04B 1/2403
8875452	12/2013	Kawai	52/167.1	F16F 7/12
10316507	12/2018	Richards	N/A	E04H 9/024
10544577	12/2019	McManus	N/A	H01H 85/08
10584477	12/2019	Richards	N/A	E04B 1/98
10669718	12/2019	Pryor	N/A	E04B 1/2403
10689876	12/2019	Richards	N/A	E04H 9/024
10760261	12/2019	Richards	N/A	E04C 3/06
11072938	12/2020	Pryor	N/A	E04H 9/024
11162260	12/2020	Pryor	N/A	E04H 9/024
11299880	12/2021	Pryor	N/A	E04B 1/24
2002/0184836	12/2001	Takeuchi	52/289	E04C 3/04
2004/0050013	12/2003	Okada	52/848	E04C 3/32
2004/0187430	12/2003	Takeuchi	52/837	E04B 5/10
2004/0237456	12/2003	Simmons	52/837	E04C 3/04

2005/0055970	12/2004	Simmons	52/837	E04B 1/2403
2006/0144006	12/2005	Suzuki	52/655.1	E04H 9/024
2006/0265992	12/2005	Hiragaki	52/633	E04B 5/43
2009/0223146	12/2008	Platt	52/745.2	E04B 9/10
2009/0223166	12/2008	Ohata	52/699	E04B 1/2403
2010/0205876	12/2009	Christopoulos	52/698	E04H 9/024
2011/0030305	12/2010	Karns	52/655.1	E04B 1/2403
2013/0074427	12/2012	Kawai	188/371	E04H 9/0237
2014/0182234	12/2013	Yang	52/655.1	E04B 1/2403
2014/0325932	12/2013	Tran	52/700	E04C 3/06
2015/0159362	12/2014	Pryor	52/655.1	E04B 1/1903
2015/0159369	12/2014	Chen	52/167.8	E04B 1/2403
2015/0275501	12/2014	Houghton	52/854	E04H 9/024
2017/0218616	12/2016	Richards	N/A	E04H 9/024
2018/0274223	12/2017	Richards	N/A	E04C 3/06
2018/0347222	12/2017	Richards	N/A	E04H 9/021
2019/0186165	12/2018	Pall	N/A	E04B 1/92
2019/0249410	12/2018	Richards	N/A	E04H 9/024
2019/0292783	12/2018	Pryor	N/A	E04C 3/38
2020/0011051	12/2019	Hou	N/A	E04G 23/02
2020/0362585	12/2019	Hou	N/A	E04H 9/021
2022/0316202	12/2021	Richards	N/A	E04H 9/024
2022/0333369	12/2021	Richards	N/A	E04B 1/2403
2022/0333397	12/2021	Richards	N/A	E04B 1/2403
2023/0058224	12/2022	Evans	N/A	E04B 1/2608
2023/0175248	12/2022	Richards	52/167.1	E04B 1/2403

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101344813	12/2012	KR	N/A
101406485	12/2013	KR	N/A
20140105643	12/2013	KR	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 63/287,308, filed Dec. 8, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims priority to U.S. Provisional Patent Application No. 63/287,308 filed on Dec. 8, 2021, the disclosure of which is incorporated herein, in its entirety, by this reference.

BACKGROUND

(1) When designing structures to resist severe earthquake or wind loads, engineers may rely on ductility to prevent catastrophic failure. Engineers may design certain parts of the building to yield in a controlled manner in order to accommodate the large movements associated with severe

earthquakes and wind loads. The parts of the structure that are typically designed to yield in a controlled manner are beams, braces, walls, and/or columns.

SUMMARY

(2) Embodiments are directed to structural fuses, structures including the same, and methods of using and forming the same. In an embodiment, a structural fuse is disclosed. The structural fuse includes a first attachment region defining one or more first bolt holes, a second attachment region defining one or more second bolt holes, and a yielding region between the first attachment region and the second attachment region. The yielding region defines one or more slots exhibiting an elongated shape. The yielding region is configured to be slidably attached to a structural element via the one or more slots.

(3) In an embodiment, a structure is disclosed. The structure includes a first structural element, a second structural element spaced from the first structural element by a gap, and at least one structural fuse attaching the first structural element to the second structural element. The structural fuse includes a first attachment region attached to the first structural element, a second attachment region attached to the second structural element, and a yielding region between the first attachment region and the second attachment region. The yielding region slidably attached to the second structural element.

(4) Features from any of the disclosed embodiments may be used in combination with one another, without limitation. In addition, other features and advantages of the present disclosure will become apparent to those of ordinary skill in the art through consideration of the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings illustrate several embodiments of the present disclosure, wherein identical reference numerals refer to identical or similar elements or features in different views or embodiments shown in the drawings.

(2) FIGS. 1A and 1B are a bottom plan view and a side elevational view of a structure, according to an embodiment.

(3) FIG. 1C is a bottom plan view of the structure when an external compressive load is applied to the structure that is sufficiently large to yield the structural fuse, according to an embodiment.

(4) FIG. 2 is a bottom plan view of a structural fuse, according to an embodiment.

(5) FIGS. 3A and 3B are side elevational and bottom plan views, respectively, of a structure that includes one or more straps moveably attaching a yielding region of a structural fuse to a second structural element, according to an embodiment.

(6) FIGS. 4A-4C are side elevational, bottom plan, and top plan views, respectively, of a structure, according to an embodiment.

(7) FIGS. 5A and 5B are side elevational and bottom plan views, respectively, of a structure, according to an embodiment.

(8) FIG. 6 is a side elevational view of a structure, according to an embodiment.

(9) FIGS. 7A and 7B are a side elevational and front elevational view of a structure, according to an embodiment.

(10) FIG. 8A is a side elevational view of a structure, according to an embodiment.

(11) FIG. 8B is a cross-sectional view of the structure taken along plane 8B-8B.

DETAILED DESCRIPTION

(12) Embodiments are directed to structural fuses, structures including the same, and methods of using and forming the same. An example structural fuse includes a first attachment region configured to be attached to a first structural element, a second attachment region configured to be

attached to a second structural element, and a yielding region extending between the first and second attachment regions. The yielding region is configured to be slidably attached to the second structural element.

(13) The structural fuse may form part of a structure. The structure includes a first structural element and a second structural element. The first and second structural elements may include a column, a beam, a link beam, a coupling beam, a brace, a wall, a floor, or foundation. The first and second attachment regions of the structural fuse may be rigidly attached (e.g., using bolts, rivets, welds, nails, etc.) to the first and second structural elements, respectively. The yielding region may be slidably attached to the second structural element. During loading (e.g., caused by earthquakes or wind), the portions of the first and second structural elements may move closer together or further apart. The movement of the first and second structural elements causes the first and second attachment regions of the structural fuse to likewise move due to the rigid attachments between the first and second structural elements and the first and second attachment regions, respectively, of the structural fuse. The attachment between the yielding region and the second structural element allows the yielding region to move relative to the first and second structural elements. The movement of the first and second attachment regions of the structural fuse caused by the movement of the first and second structural elements may cause a tensile or compressive stress to be applied to the yielding region of the structural fuse which may cause the yielding region to yield. The yielding region of the structural fuse may want to buckle when a sufficiently large compressive stress is applied thereto. However, the slidable connection between the yielding region of the structural fuse and the second structural element may mitigate or restrain buckling of the yielding region of the structural fuse thereby preventing the undesirable drop in strength in the structural fuse caused by such buckling.

(14) The structural fuses disclosed herein exhibit one or more improvements over conventional yielding elements (i.e., elements in conventional structures that are designed to yield in a controlled manner to accommodate movements caused by earthquakes and wind loads). In an example, the conventional yielding elements includes a structural element, such as a column, beam, a link beam, a brace (e.g., a buckling-restrained brace), a wall, a floor, or foundation. In such an example, yielding the structural requires replacement of the entire structural element. Replacing the entire yielded structural element may be costly and difficult. In some instances, replacing the entire yielded structural element may be impossible due to cost or due to weakening the structure. Meanwhile, the structural fuses disclosed herein are configured to yield before the structural elements to which the structural fuses are attached. It is significantly easier and simpler to replace the structural fuse than the structural element. In an example, the conventional yielding elements may include a Simpson yield link. The Simpson yield link includes a Simpson structural fuse which is different from the structural fuses disclosed herein, an additional plate covering an exterior of the Simpson structural fuse, and one or more fillers that are configured to be attached to structural elements. The Simpson yield link is more complex to attach to the structural elements than the structural fuses disclosed herein due to the inclusion of the additional plate and the one or more fillers. The increased complexity of the Simpson yield link increases the likelihood that the Simpson yield link is incorrectly installed and increases the time required to install the Simpson yield link compared to the structural fuses disclosed herein.

(15) FIGS. 1A and 1B are a bottom plan view and a side elevational view of a structure **100**, according to an embodiment. The structure **100** includes a first structural element **102** and a second structural element **104**. The first and second structural elements **102**, **104** are spaced from each other by a gap **106**. The structure **100** also has a structural fuse **108** that is attached to the first and second structural elements **102**, **104**. The structural fuse **108** is configured to attach the first and second structural elements **102**, **104** together. The structural fuse **108** is also configured to preferentially yield when the first and second structural elements **102**, **104** move relative to each other thereby preventing or at least inhibiting yielding of the first and second structural elements

102, 104. Yielding the structural fuse **108** instead of the first and second structural elements **102, 104** may make repair of the structure **100** significantly easier. The structural fuse **108** is also configured to prevent or at least inhibit buckling thereof when the structural fuse **108** yields due to a compressive load.

(16) The first and second structural elements **102, 104** may include any elements that may form part of the structure **100**. In an example, as illustrated, the first and second structural elements **102, 104** may be plates. In an example, the first and second structural elements **102, 104** may include an I-beam, a hollow structural section (e.g., a square, rectangular, or circular hollow structural section), or other structural element. In an example, the first and second structural elements **102, 104** may be aligned generally parallel (as shown) or perpendicular (as shown in FIG. 7A) to each other. In an example, the first and second structural elements **102, 104** may be a column, a beam, a link beam, a coupling beam, a brace, a wall, a floor, a foundation, or any other structural element.

(17) The structural fuse **108** includes a first attachment region **110** and a second attachment region **112**. The first and second attachment regions **110, 112** are spaced from each other, for example, along a longitudinal axis **114** of the structural fuse **108**. For example, the first and second attachment regions **110, 112** may be spaced from each other by a yielding region **116**. The first attachment region **110** is configured to be attached, either directly (as shown) or indirectly (as shown in FIG. 4B) to the first structural element **102**. For example, the first attachment region **110** is attached to a portion of the first structural element **102** that is adjacent or proximate to a terminal end **118** of the first structural element **102**. The second attachment region **112** is configured to be attached, either directly (as shown) or indirectly, to a portion of the second structural element **104** that is adjacent to, proximate to, or spaced from a terminal end **120** of the second structural element **104**. It is noted that the first attachment region **110** and/or the second attachment region **112** may be attached to a portion of the first structural element **102** or the second structural element **104**, respectively, that is spaced from the terminal end of the structural element, as shown in FIG. 4A.

(18) The first and second attachment regions **110, 112** may be rigidly (i.e., non-movably) attached to the first and second structural elements **102, 104** using any suitable technique. In an embodiment, as illustrated, the first and second attachment regions **110, 112** may be rigidly attached to the first and second structural elements **102, 104** using one or more bolts **122** (as shown) or one or more rivets. In such an embodiment, the first structural element **102**, the second structural element **104**, the first attachment region **110**, and second attachment region **112** may define one or more bolt holes **124** (shown using dashed lines in FIG. 1B). The bolt holes **124** of the first structural element **102** and the bolt holes **124** of the first attachment region **110** may be aligned with each other and the bolt holes **124** of the second structural element **104** and the bolt holes **124** of the second attachment region **112** may be aligned with each other. The bolt holes **124** of the first structural element **102**, the second structural element **104**, the first attachment region **110**, and the second attachment region **112** may be aligned in one or more rows that are generally parallel to the longitudinal axis **114** of the structure **100**. For example, as illustrated, the bolt holes **124** may be aligned in two rows which, for example, may allow each row of bolt holes **124** to extend through a flange with an intersecting web therebetween when the first and second structural elements **102, 104** are I-beams. However, it is noted that the bolt holes **124** may be arranged in a single row or three or more rows.

(19) In an embodiment, at least one of the first structural element **102** and the first attachment region **110** or the second structural element **104** and the second attachment region **112** may be attached together using a technique other than or in addition to the one or more bolts **122** or one or more rivets. For example, at least one of the first structural element **102** and the first attachment region **110** or the second structural element **104** and the second attachment region **112** may be attached together using a weld, clamps, straps, or any other suitable technique.

(20) The structural fuse **108** is configured to form a gap **106** between the first and second structural elements **102, 104** when the first and second attachment regions **110, 112** are attached to the first

and second structural elements **102**, **104**, respectively, and substantially no external load (e.g., caused by earthquakes or wind loads) is applied to the structure **100**. The gap **106** allows the first and second structural elements **102**, **104** to move closer to each other without contacting each other when a compressive load is applied to the structural fuse **108**. Without the gap **106**, the first and second structural elements **102**, **104** may need to deform to accommodate the compressive load instead of or in addition to yielding the structural fuse **108**. The gap **106** may be selected to be about 0.5 cm to about 1 cm, about 0.75 cm to about 1.25 cm, about 1 cm to about 1.5 cm, about 1.25 cm to about 1.75 cm, about 1.5 cm to about 2 cm, about 1.75 cm to about 2.25 cm, about 2 cm to about 2.5 cm, about 2.25 cm to about 2.75 cm, about 2.5 cm to about 3 cm, about 2.75 cm to about 3.25 cm, about 3 cm to about 3.5 cm, about 3.25 cm to about 3.75 cm, about 3.5 cm to about 4 cm, about 3.75 cm to about 4.5 cm, about 4 cm to about 5 cm, about 4.5 cm to about 5.5 cm, about 5 cm to about 6 cm, about 5.5 cm to about 6.5 cm, about 6 cm to about 7 cm, about 6.5 cm to about 8 cm, about 7 cm to about 9 cm, about 8 cm to about 10 cm, about 9 cm to about 11 cm, about 10 cm to about 12 cm, about 11 cm to about 13 cm, about 12 cm to about 14 cm, or about 13 cm to about 15 cm. The gap **106** may be selected based on the amount of deformation that the structure **100** is configured to accommodate during severe loading. The amount of deformation that the structure **100** is configured to accommodate during severe loading may vary based on the type of structure. In an example, a structure that includes a beam-to-column connection (as shown in FIGS. 4A-4C) may include a gap of about 0.5 cm to about 5 cm or about 1.5 cm to about 3.5 cm. In an example, a structure that is a brace (as shown in FIGS. 8A and 8B) may include a gap that is greater than 3 cm, such as about 6 cm to about 10 cm.

(21) The structural fuse **108** includes a yielding region **116** positioned between the first attachment region **110** and the second attachment region **112**. The yielding region **116** is configured to preferentially yield when a sufficiently large load is applied to the structure **100**. In particular, the yielding region **116** is configured to yield before the first structural element **102**, the second structural element **104**, the first attachment region **110**, and the second attachment region **112** yields thereby preventing the need to repair a yielded first or second structural element **102**, **104** and preventing weakening of the attachment between the first and second structural elements **102**, **104** and the structural fuse **108**.

(22) The yielding region **116** of the structural fuse **108** may include one or more weakening features that are configured to weaken the yielding region **116** thereby causing the yielding region **116** to preferentially yield before the first structural element **102**, the second structural element **104**, the first attachment region **110**, and the second attachment region **112** yields. In an embodiment, as illustrated in FIG. 1B, the weakening features of the yielding region **116** includes one or more cutouts **128** formed in the yielding region **116**. In an example, the cutouts **128** extend inwardly from an outermost peripheral edge **130** of the structural fuse **108**. The cutouts **128** cause the yielding region **116** to exhibit a width $W_{sub.Y}$ that is less than the width $W_{sub.F}$ of the first attachment region **110** and less than the width $W_{sub.S}$ of the second attachment region **112**, wherein the widths $W_{sub.Y}$, $W_{sub.F}$, and $W_{sub.S}$ are measured perpendicular to the longitudinal axis **114** of the structural fuse **108**. Assuming the thickness of the structural fuse **108** is substantially constant (e.g., the structural fuse **108** is formed from a plate), the smaller width $W_{sub.Y}$ causes the yielding region **116** to exhibit larger stresses than first and second attachment regions **110**, **112** thereby causing the yielding region **116** to preferentially yield.

(23) In an example, the cutouts **128** may be configured to cause the width $W_{sub.Y}$ of the yielding region **116** to be smaller than at least one of the width $W_{sub.F}$ of the first attachment region **110** or the width $W_{sub.S}$ of the second attachment region **112** by about 0.25 cm or more, about 0.5 cm or more, about 0.75 cm or more, about 1 cm or more, about 1.25 cm or more, about 1.5 cm or more, about 2 cm or more, about 2.5 cm or more, about 3 cm or more, about 3.5 cm or more, about 4 cm or more, about 5 cm or more, about 6 cm or more, about 7 cm or more, about 8 cm or more, about 9 cm or more, about 12.5 cm or more, about 15 cm or more, about 17.5 cm or more, about 20 cm or

more, about 25 cm or more, about 30 cm or more, or in ranges of about 0.25 cm to about 0.75 cm, about 0.5 cm to about 1 cm, about 0.75 cm or about 1.25 cm, about 1 cm to about 1.5 cm, about 1.25 cm to about 1.75 cm, about 1.5 cm to about 2 cm, about 1.75 cm to about 2.5 cm, about 2 cm to about 3 cm, about 2.5 cm to about 3.5 cm, about 3 cm to about 4 cm, about 3.5 cm to about 5 cm, about 4 cm to about 6 cm, about 5 cm to about 7 cm, about 6 cm to about 8 cm, about 7 cm to about 9 cm, about 8 cm to about 10 cm, about 9 cm to about 12.5 cm, about 10 cm to about 15 cm, about 12.5 cm to about 17.5 cm, about 15 cm to about 20 cm, about 17.5 cm to about 25 cm, or about 20 cm to about 30 cm. In an example, the cutouts **128** may be configured to cause the width $W_{sub.Y}$ of the yielding region **116** to be smaller than at least one of the width $W_{sub.F}$ of the first attachment region **110** or the width $W_{sub.S}$ of the second attachment region **112** by about 1% to about 5%, about 2.5% to about 7.5%, about 5% to about 10%, about 7.5% to about 15%, about 10% to about 20%, about 15% to about 25%, about 20% to about 30%, about 25% to about 35%, about 30% to about 40%, about 35% to about 45%, about 40% to about 50%, about 45% to about 55%, about 50% to about 60%, about 55% to about 65%, about 60% to about 70%, about 65% to about 75%, about 70% to about 80%, or about 75% to about 85%. The difference between the width $W_{sub.Y}$ of the yielding region **116** and the widths $W_{sub.F}$, $W_{sub.S}$ of the first and second attachment regions **110**, **112** may depend on the size of the structure **100**, the maximum expected movement between the first and second structural elements **102**, **104** when a severe load is applied to the structure **100**, and the desired load at which the yielding region **116** yields.

(24) In an example, as shown, any corners formed by the cutout **128** are rounded (e.g., exhibit an average radius of curvature that is greater than 0.25 cm). The rounded corners of the cutout **128** prevent or at least inhibit the formation of stress concentrators which may cause the yielding region **116** to prematurely yield. Further, corners formed by the cutout **128** are typically close to the first and second attachment regions **110**, **112** and such stress concentrators may cause the first and second attachment regions **110**, **112** to yield instead of or in addition to the yielding region **116**. In an example, any corners formed by the cutout **128** are not rounded.

(25) The structural fuse **108** may include weakening features other than or in addition to the cutouts **128**. Examples of other weakening features includes selectively thinning the yielding region **116** relative to the first and second attachment regions **110**, **112** and forming the yielding region **116** from a weaker material (i.e., a material exhibiting a yielding stress) that is less than a material forming the first and second attachment regions **112**, **112**.

(26) In an embodiment, the yielding region **116** may include one or more internal cutouts formed therein that are spaced from the peripheral edges of the **130** of the structural fuse **108**. It is noted that such internal cutouts are distinguishable from the slot **132** (discussed below) in that such internal cutouts are not configured to receive bolts **112** or rivets and/or do not include bolts **122** or rivets positioned therein.

(27) The structure **100** is configured to prevent the yielding region **116** from buckling when a compressive load is applied to the yielding region **116**. For example, allowing the yielding region **116** to buckle significantly decreases a load that causes the yielding region **116** to yield than if the yielding region **116** is prevented from buckling. Further, allowing the yielding region **116** to buckle significantly weakens the structure **100**. In an embodiment, as illustrated, the yielding region **116** is attached to the second structural element **104** using bolts **134**. The bolts **134** provide discrete (partial) out-of-plane restraint thereabout that prevents the yielding region **116** from buckling. It has been found that the discrete out-of-plane restraint provided by the bolts **134** is sufficient to prevent buckling of the yielding region **116**. It is noted that the structure **100** may include rivets or other similar element instead of or in addition to the bolts **134**.

(28) The discrete out-of-plane restraint provided by the bolts **134** is different from conventional buckling restraint systems since conventional buckling restraint systems rely on continuous out-of-plane restraint to prevent buckling. In an example, a conventional buckling restraint system includes a buckling restraint brace. The buckling restrained brace relies on concrete, motor, cement,

or other similar material (collectively referred to as “concrete”) disposed between the outer shell and the inner brace to provide continuous out-of-plane restraint to prevent buckling of the brace. However, the concrete or other filler causes the buckling restrained brace to exhibit exceptionally high weight which, in turn, requires strengthening the structure including the buckling restrained brace to accommodate the additional weight. The structures disclosed herein (e.g., structures including a structural fuse) do not require concrete or other filler to prevent buckling of the yielding region **116**. In an example, a conventional buckling restrained system includes a Simpson yield link. The Simpson yield link requires the use of an additional plate adjacent to a yielding element to provide continuous out-of-plane buckling of the yielding element. However, the additional plate of the Simpson yielding link increases the complexity of installing the Simpson yield link. However, the structures disclosed herein (e.g., structures including a structural fuse) may not include the additional plate and/or the filler element thereby making the structures disclosed herein less complex and more likely to be installed correctly.

(29) The yielding region **116** defines one or more slots **132**. The slots **132** allow the yielding region **116** to be attached to the second element **104** with the bolts **134** and allows the bolts **134** to prevent buckling of the yielding region **116**. For example, the yielding region **116** allows the shaft of the bolts **134** to pass through the structural fuse **108** and be attached to the second structural element **104**. However, the yielding region **116** exhibits a size that does not permit the head of the bolt **134** to pass therethrough thereby allowing the head of the bolt **134** to provide discrete out-of-plane restraint.

(30) The slots **132** are configured to allow the bolts **134** to move relative to the yielding region **116**. In particular, the slots **132** exhibit a size that is greater than and/or shape that is different than the shaft of the bolt **134** which permits the shaft to move in the slot **132**. In an example, the slots **132** may exhibit a generally elongated shape that extends generally parallel to the longitudinal axis **114** of the structural fuse **108**. In such an example, the elongated shape of the slots **132** allows the shafts of the bolts **134** to move in a direction that is generally parallel to the longitudinal axis **114** relative to the yielding region **116**. Examples of elongated shapes that the slots **132** may exhibit include generally oval, oblong, elliptical, rectangular, or rectangular racetrack.

(31) Allowing the shafts of the bolts **134** to move relative to the yielding region **116** also allows the first and second structural elements **102**, **104** to move relative to the yielding region **116**. Allowing the first and second structural elements **102**, **104** to move relative to the yielding region **116** causes the yielding region **116** to absorb at least some of the energy applied to the structure **100** (e.g., energy caused by earthquakes and wind). It also allows the yielding region **116** to shorten or lengthen depending on the movement of the first structural element **102** relative to the second structural element **104**.

(32) It is noted that the slots **132** exhibit a size that is sufficiently large that the shaft of the bolts **134** are unlikely to contact opposing portions of the slots **132** that are spaced from each other in a direction that is parallel to the longitudinal axis **114**. For example, allowing the shafts to contact such opposing portions of the slots **132** prevents further movement of the first and second structural elements **102**, **104** in one direction relative to the yielding region **116**. Preventing further movement of the first and second structural elements **102**, **104** relative to the yielding region **116** effectively makes the attachment between the second structural element **104** and the yielding region **116** a rigid attachment. This rigid attachment between the second structural element **104** and the yielding region **116** limits the ability of the yielding region **116** to absorb energy applied to the structure **100**, increases the likelihood that the first and second structural elements **102**, **104** yield, and increases the likelihood that the first and second attachment regions **110**, **112** yield.

(33) It is noted that the bolts **134** may prevent at least some movement between second structural element **104** and the yielding region **116** due to static friction between the yielding region **116** and the second structural element **104** and static friction between the bolts **134** and the yielding region **116**. This restriction of movement caused by the static friction is considered negligible because the

static friction is overcome by a load that is less than the load required to yield any one of the first structural element **102**, the second structural element **104**, the first attachment region **110**, the second attachment region **112**, the rigid attachment between the first structural element **102** and the first attachment region **110**, and the rigid attachment between the second structural element **104** and the second attachment region **112**. In an example, the static friction between the second structural element **104**, the yielding region **116**, and the bolt **134** may be reduced by loosening the bolt **134** relative to the bolts **122**. In an example, the static friction between the second structural element **104**, the yielding region **116**, and the bolts **134** may be reduced by applying lube (e.g., oil, grease, graphite) between at least one of the second structural element **104** and the yielding region **116** or the yielding region **116** and the head of the bolt **134**.

(34) The yielding region **116** may be configured to yield when a tensile or compressive load and not a shear load is applied thereto. In an example, the yielding region **116** may be configured to yield when a tensile or compressive load is applied thereto when the slots **132** are generally centrally aligned with the longitudinal axis **114**. Generally centrally aligning the slots **132** with the longitudinal axis **114** ensures that the yielding region **116** yields when a tensile or compressive load is applied thereto. In an example, when the structural fuse **108** defines bolt holes **124**, the slots **132** are arranged in one or more rows that are aligned with the one or more rows of the bolt holes **124**. For instance, in the illustrated embodiment, the slots **132** are aligned in two rows that are aligned with and parallel to the two rows of bolt holes **124**. In an example, the slots **132** exhibit an elongated shape and the elongated shape is generally parallel to the longitudinal axis **114** and/or aligned with the one or more rows of bolt holes **124**. Aligning the elongated shape of the slots **132** with the longitudinal axis **114** ensures that the yielding region **116** yields when a tensile or compressive load is applied thereto.

(35) FIGS. **1B** and **1C** illustrate how the structural fuse **108** absorbs energy when a load is applied to the structure **100**. FIG. **1B** illustrates the structure **100** when no external load is applied to the structure **100**. When no external load is applied to the structure **100**, the first and second structural elements **102**, **104** are spaced from each other by the gap **106** (illustrated in FIG. **1B**, partially obscured in FIG. **1A**) and the bolts **134** are generally centrally located in the slots **132**. FIG. **1C** is a bottom plan view of the structure **100** when an external compressive load F is applied to the structure **100** that is sufficiently large to yield the structural fuse **108**, according to an embodiment. The compressive load F causes the first and second structural elements **102**, **104** to move closer together. Moving the first and second structural elements **102**, **104** closer together also causes the first and second attachment regions **110**, **112** of the structural fuse **108** to move closer together due to the rigid attachment between the first structural element **102** and the first attachment region **110** and the rigid attachment between the second structural element **104** and the second attachment region **112**. Moving the first and second attachment regions **110**, **112** closer together may be sufficient to cause the yielding region **116** to yield, as shown in FIG. **1C** using cross-hatching. Yielding the yielding region **116** causes the structural fuse **108** to absorb at least some of the energy applied to the structure **100**. Also, the second structural element **104** is able to move relative to the yielding region **116** (as indicated by the bolts **134** moving the left side of the slots **132**) thereby allowing the movement of the second structural element **104** to absorb at least some of the energy and minimizing loads being applied from the yielding region **116** to the second structural element **104**. The out-of-plane restraint provided by the bolts **134** prevents the yielding region **116** from buckling thereby increasing the compressive load F required to yield the yielding region **116**.

(36) Although FIG. **1C** illustrates a compressive load F being applied to the structure **100**, it is noted that a tensile load may be applied to the structure **100**. The tensile load causes the first and second structural elements **102**, **104** to move further apart. Moving the first and second structural elements **102**, **104** apart also causes the first and second attachment regions **110**, **112** of the structural fuse **108** to move further apart due to the rigid attachment between the first structural element **102** and the first attachment region **110** and the rigid attachment between the second

structural element **104** and the second attachment region **112**. Moving the first and second attachment regions **110**, **112** apart may be sufficient to cause the yielding region **116** to yield. Yielding the yielding region **116** causes the structural fuse **108** to absorb at least some of the energy applied to the structure **100**. Also, the second structural element **104** is able to move relative to the yielding region **116** thereby allowing the movement of the second structural element **104** to absorb at least some of the energy and minimize loads being applied from the yielding region **116** to the second structural element **104**.

(37) The structures disclosed herein may include structural fuses that are different than the structural fuse **108** illustrated in FIGS. **1A-1C**. For example, FIG. **2** is a bottom plan view of a structural fuse **208**, according to an embodiment. Except as otherwise disclosed herein, the structural fuse **208** is the same or substantially similar to any of the structural fuses disclosed herein. For example, the structural fuse **208** includes a first attachment region **210** that is configured to be rigidly attached to a first structural element (not shown), a second attachment region **212** that is configured to be rigidly attached to the second structural element (not shown), and a yielding region **216** extending between the first and second attachment regions **210**, **212**.

(38) The yielding region **216** includes a plurality of slots **232**. In an example, as illustrated, the plurality of slots **232** may be arranged in two or more rows. Each of the rows may extend generally parallel to a longitudinal axis **214** of the structural fuse **208**. The plurality of slots **232** may also be arranged in one or more columns. The columns may extend generally perpendicular to the longitudinal axis **214** of the structural fuse **208**. Arranging the slots **232** in the two or more rows and the one or more columns causes any loads applied to the structural fuse **208** to be generally uniformly applied to the yielding region **216**. Generally uniformly applying the loads to the yielding region **216** promotes distributed yielding in the yielding region **216**.

(39) In an embodiment, not shown, each of the slots **232** includes a bolt **234** partially positioned therethrough. In an embodiment, as shown, only some of the slots **232** includes a bolt **234** partially positioned therethrough. It has been found that positioning the bolts **234** in some of the slots **232** (e.g., in only one slot **232** in each column and/or in every other slot **232**) may provide sufficient out-of-plane restraint to prevent or at least inhibit buckling of the yielding region **216**.

(40) As previously discussed, the yielding region of the structural fuses disclosed herein may be moveably attached to the second structural element using one or more techniques instead of or in addition to the bolts. For example, FIGS. **3A** and **3B** are side elevational and bottom plan views, respectively, of a structure **300** that includes one or more straps **334** moveably attaching a yielding region **316** of a structural fuse **308** to a second structural element **304**, according to an embodiment. Except as otherwise disclosed herein, the structure **300** is the same or substantially similar to any of the structures disclosed herein. For example, the structure **300** includes a first structural element **302**, a second structural element **304**, and a structural fuse **308**. The structural fuse **308** includes a first attachment region **310** rigidly attached (e.g., via a weld **335**, shown using a bolded line) to the first structural element **302**, a second attachment region **312** rigidly attached (e.g., via a weld **335**, shown using a bolded line) to the second structural element **304**, and a yielding region **316** between the first and second attachment regions **310**, **312**.

(41) The yielding region **316** is moveably attached to the second structural element **304** using one or more straps **334** (e.g., one or more metal straps). The straps **334** extend at least partially around the second structural element **304** and the yielding region **316** thereby moveably attaching the second structural element **304** to the yielding region **316**. The straps **334** provide discrete out-of-plane restraint thereabout that prevents the yielding region **316** from buckling. It has been found that the discrete out-of-plane restraint provided by the straps **334** is sufficient to prevent buckling of the yielding region **316**. In an embodiment, the straps **334** are rigidly attached to one of the second structural element **304** or the yielding region **316** using, for example, an adhesive, a weld, or another suitable attachment technique. In an embodiment, the straps **334** are not rigidly attached to either of the second structural element **304** or the yielding region **316**.

(42) Attaching the second structural element **304** to the yielding region **316** using the one or more straps **334** may facilitate manufacturing and configurability of the yielding region **316**. In an example, attaching the second structural element **304** to the yielding region **316** using the one or more straps **334** precludes the need to form one or more slots in the yielding region **316**. The lack of slots formed in the yielding region may allow a load to be more uniformly applied to the yielding region than if the yielding region **316** included slots and prevents any weakening of the yielding region caused by the slots.

(43) As previously discussed, the structural fuses disclosed herein may be used in structures other than the structures illustrated in FIGS. **1A-1C**, **3A**, and **3B**. FIGS. **4A-8B** illustrate different structures that may include any of the structural fuses disclosed herein. Except as otherwise disclosed herein, the structures and structural fuses illustrated in FIGS. **4A-8B** are the same as or substantially similar to any of the structures and structural fuses disclosed herein, respectively. It is noted that the structures illustrated in FIGS. **4A-8B** are merely examples of structures that may include the structural fuses disclosed herein. In other words, the structural fuses disclosed herein may be used in structures other than the structures illustrated in FIGS. **1A-1C**, **3A**, **3B**, and **4A-8B**.

(44) FIGS. **4A-4C** are side elevational, bottom plan, and top plan views, respectively, of a structure **400**, according to an embodiment. The structure **400** is an example of a beam-to-column connection. The structure **400** includes a column **402** (i.e., a first structural element) and a beam **404** (i.e., a second structural element). In the illustrated embodiment, the column **402** and beam **404** are illustrated as being I-beams. However, it is noted that the column **402** and the beam **404** may include plates, hollow structural sectionals, T-beams, or any other suitable structural element.

(45) Referring to FIGS. **4A** and **4B**, the column **402** and the beam **404** are attached together using a structural fuse **408**. The structural fuse **408** includes a first attachment region **410**, a second attachment region **412**, and a yielding region **416** extending between the first and second attachment regions **410**, **412**. The first attachment region **410** is rigidly attached to the column **402**, such as to a flange **436** of the column **402**. The second attachment region **412** is rigidly attached (e.g., directly or indirectly) to the beam **404**, such as to a flange **440** of the beam **404**. The structural fuse **408** is moveably attached to the beam **404**.

(46) In an example, not shown, the first attachment region **410** is directly rigidly attached to the column **402** using, for instance, a weld or one or more bolts (e.g., the first attachment region **410** is an angle). In an example, as shown, the first attachment region **410** is indirectly rigidly attached to the column **402**, for instance, using a first (e.g., bottom) plate **438**. The first plate **438** may be attached to the column **402** (e.g., using a weld, bolts, etc.) before assembling the structure **400** in the field which may facilitate assembly of the structure **400**. In an example, the second attachment region **412** is directly or indirectly attached to the beam **404**.

(47) In an example, as illustrated, the structural fuse **408** includes one or more slot **432** and the structural fuse **408** is attached to the beam **404** using one or more bolts **434**. In an example, not shown, the structural fuse **408** is attached to the beam **404** using one or more straps, one or more clamps, one or more rivets, or using any other suitable technique.

(48) In an embodiment, as shown, the structural fuse **408** may be attached to the bottom side (e.g., the bottom flange **440**) of the beam **404**. Attaching the structural fuse **408** to the bottom side of the beam **404** may facilitate inspecting, repairing, and/or replacing of the structural fuse **408**. For example, floors are typically positioned on the top side (e.g., top flange **442**) of the beam **404**. The floors may include concrete or other elements that make accessing the top side of the beam **404** difficult. Meanwhile, a ceiling is generally positioned adjacent to the bottom side of the beam **404**. The ceiling may include drop ceiling tiles or include another material that is easier to remove than the floor. Thus, attaching the structural fuse **408** to the bottom side of the beam **404** makes accessing the structural fuse **408** easier which facilitate inspecting, repairing, and replacing the structural fuse **408**.

(49) Referring to FIGS. **4A** and **4C**, the top side (e.g., top flange **442**) of the second structural

element **404** may be attached to the column **402**. In an embodiment, as illustrated, the top side of the beam **404** is attached to the column **402** using a second (e.g., top) plate **444** or another element other than a structural fuse **408** that is not configured to preferentially yield like the structural fuse **408**. As previously discussed, the top side of the beam **404** may be difficult to access and thus, attaching the top side of the beam **404** using an element other than a structural fuse **408** reduces the likelihood that the top side of the beam **404** needs to be accessed to repair the structure **400**. In an embodiment, the top side of the beam **404** may be attached to the column **402** using a structural fuse. In an embodiment, the flange **436** of the column **402** may be attached to the web **446** of the beam **404** using a shear tab **448**.

(50) Due to the structural fuse **408**, the structure **400** exhibits adequate stiffness and is able to accommodate large deformations without losing strength. The structure **400** may also be a fully restrained connection since the first plate **438** and the second plate **444** may be welded to the column **402**. The structural fuse **408** also allows the structure **400** to be more easily repaired compared to other conventional beam-to-column connections.

(51) FIGS. 5A and 5B are side elevational and bottom plan views, respectively, of a structure **500**, according to an embodiment. The structure **500** is an eccentrically braced frame. The structure **500** includes a column **502** (i.e., a first structural element), a beam **504** (i.e., a second structural element), a link beam **550** (i.e., a third structural element), and a brace **552** (i.e., a fourth structural element). The brace **552** is attached to the rest of the structure **500** (i.e., the beam **504**) at some distance from the column **502** and the link beam **550** extends between the beam **504** to the column **502**. It is noted that the beam **504** may extend pass the brace **552** thereby providing a location for the structural fuse **508** to be attached.

(52) The structure **500** includes at least one structural fuse **508**. The at least one structural fuse **508** may attach the column **502**, the beam **504**, and link beam **550** together. The at least one structural fuse **508** may also maintain a first gap **506a** between the column **502** and the link beam **550** when no external load is applied to the structure **500** thereby allowing movement between the column **502** and the link beam **550** when a load is applied to the structure **500**. In the illustrated embodiment, the first gap **506a** is between the web **546** of the link beam **550** and the flange **536** of the column **502** and between the flange **540** of the link beam **550** and the plate **538** when the structural fuse **508** is indirectly attached to the column **502** using the plate **538**. The at least one structural fuse **508** may also maintain a second gap **506b** between the beam **504** and the link beam **550** when no external load is applied to the structure **500** thereby allowing movement between the beam **504** and the link beam **550** when a load is applied to the structure **500**.

(53) In an embodiment, as shown, the structure **500** includes a single structural fuse **508** that is configured to be attached to each of the column **502**, the beam **504**, and the link beam **506**. In such an embodiment, the structural fuse **508** includes a first attachment region **510** that is configured to be rigidly attached to the column **502** (e.g., via a plate **538**), a second attachment region **512** that is configured to be rigidly attached to the beam **504**, and a third attachment region **554** positioned between the first and second attachment regions **510**, **512** that is configured to be rigidly attached to the link beam **550**. It is noted that the third attachment region **554** may be the same or substantially similar to any of the attachment regions disclosed herein. The structural fuse **508** includes a first yielding region **516a** extending between the first and third attachment regions **510**, **554** and a second yielding region **516b** extending between the second and third attachment regions **512**, **554**. The first and second yielding regions **516a**, **516b** may be moveably attached to the link beam **550** using any of the techniques disclosed herein. In an embodiment, not shown, the structure **500** may include two or more structural fuses that attach the column **502**, the beam **504**, and the link beam **550** together. For example, in such an embodiment, the structure **500** may include a first structural fuse that attaches the column **502** to the link beam **550** and a second structural fuse that attaches the beam **504** to the link beam **550**. It is noted that using only a single structural fuse **508** to attach each of the column **502**, the beam **504**, and the link beam **550** together requires less pieces

(i.e., makes assembly of the structure **500** less complex and less likely to be installed incorrectly) than if first and second structural fuses are used to attach the column **502**, the beam **504**, and the link beam **550** together. However, using a single structural fuse **508** requires replacing the whole structural fuse **508** if only one yielding region yields whereas using the first and second structural fuses allows for a more targeted repair that wastes less material.

(54) In an embodiment, the structural fuse **508** may be attached to the bottom sides of the beam **504** and the link beam **550** to facilitate replacement of the structural fuse **508** after the structural fuse **508** yields, for reasons previously discussed. In such an embodiment, one or more elements may be attached to the top sides of the beam **504** and the link beam **550** to attach the column **502**, the beam **504**, and the link beam **550** together. The one or more elements may be configured to not preferentially yield such that repair of the one or more elements is unlikely. The one or more elements may include a first top plate **544a** that attaches the column **502** to the top side of the link beam **550** and a second top plate **544b** that attaches the top side of the beam **504** to the top side of the link beam **550**. In an embodiment, the structure **500** may include a structural fuse attached to the top side of the beam **504** and the link beam **550** instead of or in addition to the structural fuse **508** attached to the bottom side of the beam **504** and the link beam **550**.

(55) The structural fuse **508** provides several advantages over conventional eccentrically braced frames. For example, the structural fuse **508** may accelerate construction of the eccentrically braced frame illustrated in FIGS. 5A and 5B compared to a fully-welded eccentrically braced frame. The structural fuse **508** may also result in better control of loads applied to the structure **500** compared to a conventional eccentrically braced frame. The structural fuse **508** further makes it easier to inspect and repair the structure **500** (e.g., after an earthquake) compared to a conventional eccentrically braced frames.

(56) FIG. 6 is a side elevational view of a structure **600**, according to an embodiment. The structure **600** is an example of a concrete shear walls or concrete steel coupled walls. The structure includes a first wall **602** (i.e., a first structural element), a second wall **604** (e.g., a second structural element), and a coupling beam **650** (i.e., a third structural element).

(57) The structure **600** includes at least one structural fuse **608**. The at least one structural fuse **608** may attach the first wall **602**, the second wall **604**, and coupling beam **650** together. The at least one structural fuse **608** may also maintain a first gap **606a** between the first wall **602** and the coupling beam **650** when no external load is applied to the structure **600** thereby allowing movement between the first wall **602** and the coupling beam **650** when a load is applied to the structure **600**. The at least one structural fuse **608** may also maintain a second gap **606b** between the second wall **604** and the coupling beam **650** when no external load is applied to the structure **600** thereby allowing movement between the second wall **604** and the coupling beam **650** when a load is applied to the structure **600**. It is noted that the first and second gaps **606a**, **606b** include the gaps between the coupling beam **650** and the plates **638a**, **638b**.

(58) In an embodiment, as shown, the structure **600** includes a single structural fuse **608** that is configured to be attached to each of the first wall **602**, the second wall **604**, and the coupling beam **650**. In such an embodiment, the structural fuse **608** includes a first attachment region **610** that is configured to be rigidly attached to the first wall **602** (e.g., via a first bottom plate **638a**), a second attachment region **612** that is configured to be rigidly attached to the second wall **604** (e.g., via a second bottom plate **638b**), and a third attachment region **654** positioned between the first and second attachment regions **610**, **612** that is configured to be rigidly attached to the coupling beam **650**. The structural fuse **608** includes a first yielding region **616a** extending between the first and third attachment regions **610**, **654** and a second yielding region **616b** extending between the second and third attachment regions **612**, **654**. The first and second yielding regions **616a**, **616b** may be moveably attached to the coupling beam **650** using any of the techniques disclosed herein. In an embodiment, not shown, the structure **600** may include two or more structural fuses that attach the first wall **602**, the second wall **604**, and the coupling beam **650** together. For example, in such an

embodiment, the structure **600** may include a first structural fuse that attaches the first wall **602** to the coupling beam **650** and a second structural fuse that attaches the second wall **604** to the coupling beam **650**. It is noted that using only a single structural fuse **608** to attach the first wall **602**, the second wall **604**, and the coupling beam **650** together requires less pieces (i.e., makes assembly of the structure **600** less complex and less likely to be installed incorrectly) than if the first and second structural fuses are used to attach the first wall **602**, the second wall **604**, and the coupling beam **650** together. However, using a single structural fuse **608** requires replacing the whole structural fuse **608** if only one yielding region yields whereas using the first and second structural fuses allows for a more targeted repair that wastes less material.

(59) In an embodiment, the structural fuse **608** may be attached to the bottom sides of the second wall **604** and the coupling beam **650** to facilitate replacement of the structural fuse **608** after the structural fuse **608** yields. In such an embodiment, one or more elements may be attached to the top sides of the second wall **604** and the coupling beam **650** to attach the first wall **602**, the second wall **604**, and the coupling beam **650** together. The one or more elements may be configured to not preferentially yield such that repair of the one or more elements is unlikely. The one or more elements may include a first top plate **644a** that attaches the first wall **602** to the top side of the coupling beam **650** and a second top plate **644b** that attaches the top side of the second wall **604** to the top side of the coupling beam **650**. In an embodiment, not shown, the structure **600** may include a structural fuse **608** attached to the top side of the second wall **604** and the coupling beam **650** instead of or in addition to the structural fuse **608** attached to the bottom side of the second wall **604** and the coupling beam **650**.

(60) The structural fuse **608** provides several advantages over conventional coupled concrete shear walls or concrete steel coupled walls. For example, the structural fuse **608** may accelerate construction of the structure **600** and be easier to construct compared to conventional coupled concrete shear walls or concrete steel coupled walls. The structural fuse **608** also results in better control of loads applied to the structure **600** compared to conventional coupled concrete shear walls or concrete steel coupled walls. The structural fuse **608** further makes it easier to inspect and repair the structure **600** (e.g., after an earthquake) compared to a conventional coupled concrete shear walls or concrete steel coupled walls.

(61) FIGS. 7A and 7B are a side elevational and front elevational view of a structure **700**, according to an embodiment. The structure **700** is an example of a framing structure or a shear wall, such as timber framing and a timber shear wall. The structure **700** includes a wall **702** (i.e., first structural element) and a floor **704** (i.e., a second structural element). It is noted that, in some embodiments, the wall **702** may be replaced with a column and/or the floor **704** may be replaced with a foundation.

(62) The structure **700** includes a structural fuse **708**. The structural fuse **708** may attach the wall **702** and the floor **704** together. The structural fuse **708** includes a first attachment region **710** that is configured to be rigidly attached to the wall **702** and a second attachment region **712** that is configured to be rigidly attached to the floor **704**. Since the wall **702** and the floor **704** extend perpendicularly to each other, the structural fuse **708** has a 90° bend at the end. The structural fuse **708** also includes a yielding region **716** extending between the first and second attachment regions **710**, **712**. The yielding region **716** may be moveably attached to the wall **702** (as shown) or the floor **704** using any of the techniques disclosed herein.

(63) FIG. 8A is a side elevational view of a structure **800**, according to an embodiment. FIG. 8B is a cross-sectional view of the structure **800** taken along plane 8B-8B. The structure **800** is an example of a buckling-restrained brace. The structure includes a first hollow structural section **802** (i.e., a first structural element), a second hollow structural section **804** (i.e., a second structural element), and a third hollow structural section **850** (i.e., a third structural element). The first and third hollow structural sections **802**, **850** are separated by a first gap **806a** that accommodates movement between the first and third hollow structural sections **802**, **850**. The second and third

hollow structural sections **804**, **850** are separated by a second gap **806b** that accommodates movement between the second and third hollow structural sections **804**, **850**. It is noted that the first and second hollow structural sections **802**, **804** may define one or more recesses **860** formed therein that facilitate attachment of the structure **800** to a larger structure.

(64) The structure **800** includes at least one structural fuse **808**. The at least one structural fuse **808** may attach the first hollow structural section **802**, the second hollow structural section **804**, and third hollow structural section **850** together. The at least one structural fuse **808** may also maintain a first gap **806a** between the first hollow structural section **802** and the third hollow structural section **850** when no external load is applied to the structure **800** thereby allowing movement between the first hollow structural section **802** and the third hollow structural section **850** when a load is applied to the structure **800**. The at least one structural fuse **808** may also maintain a second gap **806b** between the second hollow structural section **804** and the third hollow structural section **850** when no external load is applied to the structure **800** thereby allowing movement between the second hollow structural section **804** and the third hollow structural section **850** when a load is applied to the structure **800**. It is noted that the structural fuse **808** may also define the one or more recesses **860** to facilitate attachment of the structure **800** to a larger structure.

(65) In an embodiment, as shown, the structure **800** includes at least one structural fuse **808** that is configured to be attached to each of the first hollow structural section **802**, the second hollow structural section **804**, and the third hollow structural section **850**. For example, as illustrated, the structure **800** includes four structural fuses **808** (one for each surface of the hollow structural sections) that are each configured to be attached to the first hollow structural section **802**, the second hollow structural section **804**, and the third hollow structural section **850**. In such an embodiment, the structural fuse **808** includes a first attachment region **810** that is configured to be rigidly attached to the first hollow structural section **802** (e.g., via a first weld **862**, illustrated using a bolded line), a second attachment region **812** that is configured to be rigidly attached to the second hollow structural section **804** (e.g., via a second weld **864**, illustrated using a bolded line), and a third attachment region **854** positioned between the first and second attachment regions **810**, **812** that is configured to be rigidly attached to the third hollow structural section **850** (e.g., via a third weld **866**, illustrated using a bolded line). The structural fuse **808** includes a first yielding region **816a** extending between the first and third attachment regions **810**, **854** and a second yielding region **816b** extending between the second and third attachment regions **812**, **854**. The first and second yielding regions **816a**, **816b** may be moveably attached to the third hollow structural section **850** using any of the techniques disclosed herein, such as using straps **834**. It is noted that, in the illustrated embodiment, the third attachment region **850** exhibits a width that is substantially the same as the widths of the first and second yielding regions **816a**, **816b**. However, it is noted that the third attachment region **850** may exhibit a width that is greater than the widths of the first and second yielding regions **816a**, **816b**.

(66) In an embodiment, not shown, the structure **800** may include a first structural fuse that attaches the first hollow structural section **802** to the third hollow structural section **850** and a second structural fuse that attaches the second hollow structural section **804** to the third hollow structural section **850**. It is noted that using only a single structural fuse **808** to attach each of the first hollow structural section **802**, the second hollow structural section **804**, and the third hollow structural section **850** together requires less pieces (i.e., makes assembly of the structure **800** less complex and less likely to be installed incorrectly) than if the first and second structural fuses are used to attach the first hollow structural section **802**, the second hollow structural section **804**, and the third hollow structural section **850** together.

(67) The structure **800** may be easier to fabricate using an automated process than conventional buckling-restrained braces. The structural fuses **808** makes inspecting and repairing the structure **800** easier than conventional buckling-restrained braces since the structural fuses **808** are on the exterior of the structure **800** whereas the yielding elements of conventional buckling-restrained

braces are on the interior of such braces and surrounded by concrete. Also, replacing the structural fuses **808** allows the first, second, and third hollow structural sections **802**, **804**, **850** to remain in place whereas replacing the yielding element of the conventional buckling-restrained brace requires removing all of the conventional buckling-restrained brace. Further, the structure **800** may weight substantially less than a conventional buckling-restrained brace.

(68) While various aspects and embodiments have been disclosed herein, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting.

(69) Terms of degree (e.g., “about,” “substantially,” “generally,” etc.) indicate structurally or functionally insignificant variations. In an example, when the term of degree is included with a term indicating quantity, the term of degree is interpreted to mean $\pm 10\%$, $\pm 5\%$, or $\pm 2\%$ of the term indicating quantity. In an example, when the term of degree is used to modify a shape, the term of degree indicates that the shape being modified by the term of degree has the appearance of the disclosed shape. For instance, the term of degree may be used to indicate that the shape may have rounded corners instead of sharp corners, curved edges instead of straight edges, one or more protrusions extending therefrom, is oblong, is the same as the disclosed shape, etc.

Claims

1. A structural fuse, comprising: a first attachment region defining one or more first bolt holes; a second attachment region defining one or more second bolt holes; and a third attachment region defining one or more third bolt holes; a yielding region between the first attachment region and the second attachment region, the yielding region defining one or more slots exhibiting an elongated shape, the yielding region configured to be slidably attached to a structural element via the one or more slots; and an additional yielding region between the second attachment region and the third attachment region, the additional yielding region configured to be slidably attached to an additional structural element via one or more additional slots; wherein the structural fuse is a plate; wherein the yielding region is between the first attachment region and the third attachment region; and wherein the one or more first bolt holes, the one or more second bolt holes, and the one or more third bolt holes extend through the plate in a direction parallel to each other and perpendicular to a plane in which the plate extends.
2. The structural fuse of claim 1, wherein the first attachment region exhibits a first width, the second attachment region exhibits a second width, and the yielding region exhibits a third width that is less than the first width and the second width.
3. The structural fuse of claim 1, wherein the one or more third bolt holes and the one or more additional slots are aligned in the one or more rows.
4. The structural fuse of claim 1, wherein the one or more first bolt holes, the one or more second bolt holes, and the one or more slots are aligned in one or more rows that extend generally parallel to a longitudinal axis of the structural fuse.
5. A structure, comprising: a first structural element; a second structural element spaced from the first structural element by a gap; and at least one structural fuse attaching the first structural element to the second structural element, the structural fuse including: a first attachment region attached to the first structural element, the first attachment region defining one or more first bolt holes; a second attachment region attached to the second structural element, the second attachment region defining one or more second bolt holes; and a yielding region between the first attachment region and the second attachment region, the yielding region slidably attached to the second structural element and defining one or more slots, the one or more slots exhibiting an elongated shape, the yielding region attached to the second structural element using one or more bolts extending through the one or more slots; and wherein the at least one structural fuse attaching the first structural element to the second structural element only abuts at least one of a bottom surface

of the first structural element or a bottom surface of the second structural element; and wherein the one or more first bolt holes, the one or more second bolt holes, and the one or more slots are aligned with each other in one or more rows that extend generally parallel to a longitudinal axis of the structural fuse, the one or more slots arranged on two or more columns.

6. The structure of claim 5, wherein the first structural element includes a column and the second structural element includes a beam.

7. The structure of claim 5, wherein the first structural element includes a beam and the second structural element includes a link beam.

8. The structure of claim 5, wherein the first structural element includes a wall and the second structural element includes a beam.

9. The structure of claim 5, wherein: the first structural element includes at least one of a wall or column; and the second structural element includes at least one of a floor or foundation.

10. The structure of claim 5, wherein the first attachment region is attached to the first structural element indirectly using at least one or more plates.

11. The structure of claim 5, further comprising a third structural element; and wherein the at least one structural fuse includes a third attachment region attached to the third structural element and an additional yielding region between the second attachment region and the third attachment region, the additional yielding region slidably attached to the second structural element; and wherein the yielding region is between the first attachment region and the third attachment region.

12. The structure of claim 5, wherein: the first structural element and the second structural element includes a hollow structural section having four sides; and the at least one structural fuse is attached to one of the four sides of the first structural element and the second structural element.

13. The structure of claim 5, wherein the one or more slots includes a plurality of slots and only at least some of the slots arranged in a plurality of rows and a plurality of columns; and only two or more of the plurality of slots include the one or more bolts extending therethrough, the two or more of the plurality of slots selected to generally uniformly distribute yielding in the yielding region.

14. The structure of claim 5, wherein the first attachment region exhibits a first width, the second attachment region exhibits a second width, and the yielding region exhibits a third width that is less than the first width and the second width.

15. The structure of claim 14, wherein the at least one structural fuse defines a cutout having one or more corners, and wherein the one or more corners are rounded.

16. The structure of claim 5, further comprising one or more straps extending around and slidably attaching the first structural element to the yielding region.

17. The structure of claim 5, wherein the structure does not include an additional plate disposed adjacent to an exterior surface of the structural fuse that is configured to prevent buckling of the structural fuse.

18. The structure of claim 5, wherein the yield region is configured to yield when a tensile load or a compressive load is applied to the at least one structure fuse.

19. The structure of claim 5, wherein the at least one structural fuse only exhibits discrete out-of-plane buckling restraint.

20. The structure of claim 5, wherein the yielding region is directly attached to the second structural element.

21. The structure of claim 5, wherein the one or more first bolt holes, the one or more second bolt holes, and the one or more slots are aligned with each other in one or more rows.
